

TOWER VERTICALITY TESTING





GRAVITY
ELEVATING SOLUTIONS

ABOUT US

Gravity was founded in South Africa in 1999. Since then, Gravity has been a leader in the work-at-height industry, assisting in the development of a variety of standards, providing excellent training and producing innovative solutions for worksites and a range of fall arrest equipment.

Gravity is committed to providing quality services and products, with the clients' satisfaction always being the highest priority.

INDEX

A. SURVEYING

1. Theodolite Surveying	1
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B. THE THEODOLITE

1. Classification of Theodolites	2
2. Size of a Theodolite	5
3. Description of a Theodolite	6
4. Terms Used in Manipulating a Theodolite	7
5. Adjustment of a Theodolite	8
6. Errors in Theodolite	11
7. How to Use Theodolites	13

C. TOWER VERTICALITY TESTING

1. Permissible Tilt	14
2. Procedure: Testing Tower Verticality	18

A. SURVEYING

- Surveying or land surveying is the technique, profession, and science of determining the terrestrial or three-dimensional position of points and the distances and angles between them.
- Examine and record the area and features of an area of land so as to construct a map, plan, or description.

1. *Theodolite Surveying*

The system of surveying in which the angles are measured with the help of a theodolite, is called Theodolite surveying.

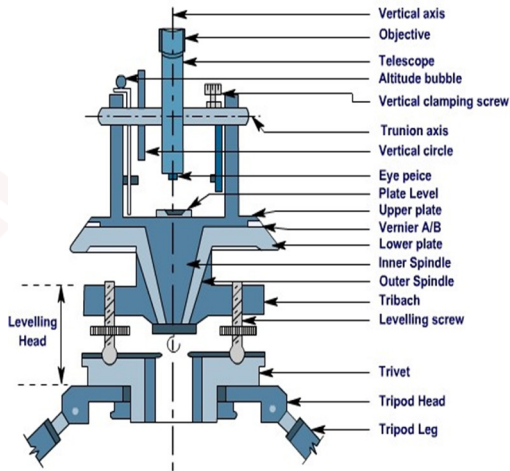
B. THE THEODOLITE

The Theodolite is a most accurate surveying instrument mainly used for:

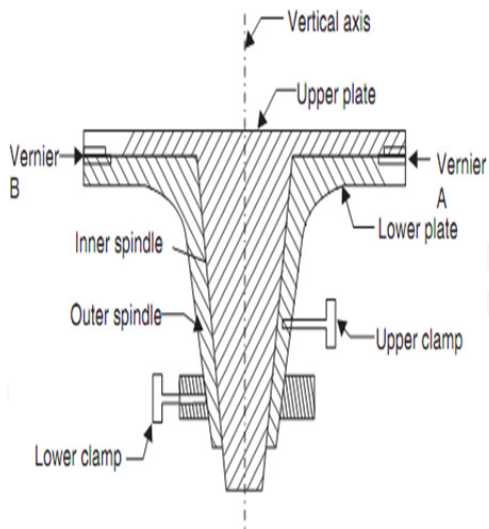
- Measuring horizontal and vertical angles.
- Locating points on a line.
- Prolonging survey lines.
- Finding difference of level.
- Setting out grades.
- Ranging curves.
- Tacheometric Survey

Theodolites may be classified as:

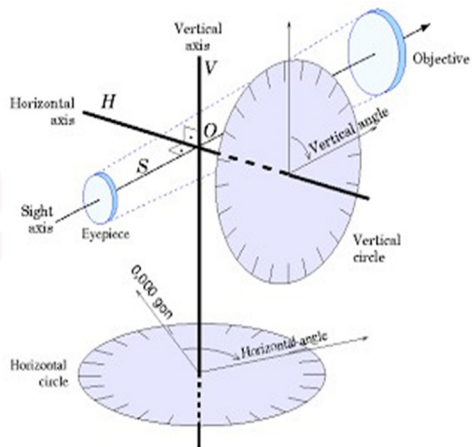
- Transit Theodolites.
- Non-Transit Theodolites.
- Vernier Theodolites.
- Micrometer Theodolites.



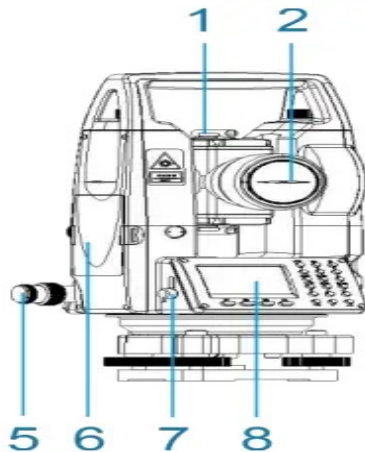
Transit Vernier Theodolite



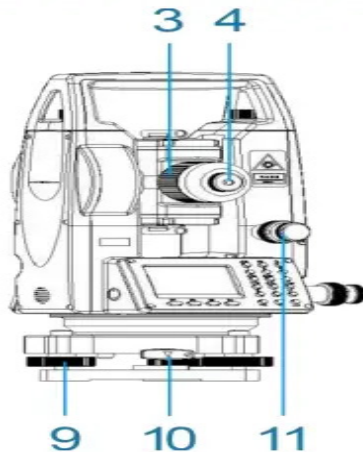
Details Of Upper And Lower Plates



Horizontal And Vertical Axis



- 1. Collimator**
- 3. Focusing Screw**
- 5. Horizontal Tangent Screw**
- 7. RS232 Interface**
- 9. Tribrach**
- 11. Vertical Tangent Screw**



- 2. Objective Lens**
- 4. Telescope Eyepiece**
- 6. Battery**
- 8. Display Unit**
- 10. Tribrach Lock**

Kolida NT-023

1. *Classification of Theodolites*

Transit Theodolite: A theodolite is called a transit theodolite when its telescope can be transited i.e., revolved through a complete revolution about its horizontal axis in the vertical plane.

Non-Transit type: The telescope cannot be transited. They are inferior in utility and have now become obsolete.

Vernier Theodolite: For reading the graduated circle if verniers are used, the theodolite is called a Vernier Theodolite.

Whereas, if a micrometer is provided to read the graduated circle, the theodolite is called a Micrometer Theodolite. Vernier type theodolites are commonly used.

2. *Size of a Theodolite*

A theodolite is designated by diameter of the graduated circle on the lower plate.

The common sizes are 8cm to 12 cm while 14 cm to 25 cm instruments are used for triangulation work.

Greater accuracy is achieved with larger theodolites as they have a bigger graduated circle with larger divisions hence used where the survey works require a high degree of accuracy.

3. *Description of a Transit Vernier Theodolite*

A Transit vernier theodolite essentially consist of the following:

1. Levelling Head
2. Lower Circular Plate
3. Upper Plate
4. Telescope
5. Vernier Scale
6. T- Frame
7. Plumb bob
8. Tripod Stand

4. Terms Used in Manipulating a Transit Vernier Theodolite

A Transit vernier theodolite essentially consist of the following:

1. Centering:

Centering means setting the theodolite exactly over an instrument-station so that its vertical axis lies immediately above the station-mark. It can be done by means of plumb bob suspended from a small hook attached to the vertical axis of the theodolite.

The centre shifting arrangement, if provided with the instrument, helps in easy and rapid performance of the centering.

2. Transiting:

Transiting is also known as plunging or reversing. It is the process of turning the telescope about its horizontal axis through 180° in the vertical plane thus bringing it upside down and making it point exactly in the opposite direction.

3. Swinging the telescope:

It means turning the telescope about its vertical axis in the horizontal plane.

A swing is called right or left according to whether the telescope is rotated clockwise or counterclockwise.

4. Face left:

If the vertical circle of the instrument is on the left side of the observer while taking a reading, the position is called the face left and the observation taken on the horizontal or vertical circle in this position, is known as the face left observation

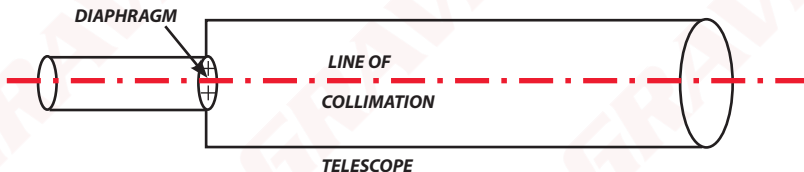
5. Face right:

If the vertical circle of the instrument is on the right side of the observer while taking a reading, the position is called the face right and the observation taken on the horizontal or vertical circle in this position, is known as the face right observation.

6. Changing face:

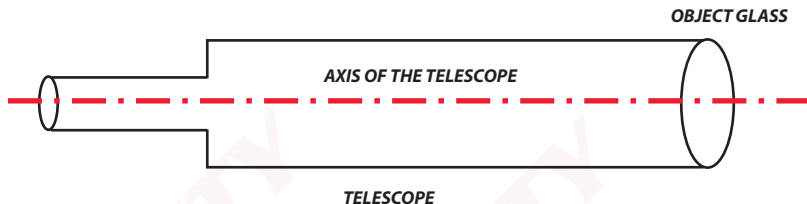
It is the operation of bringing the vertical circle to the right of the observer, if originally it is to the left, and vice – versa. It is done in two steps; Firstly, revolve the telescope through 180° in a vertical plane and then rotate it through 180° in the horizontal plane i.e., first transit the telescope and then swing it through 180° .

7. Line of collimation:



The line of collimation (LOC) is also known as the line of sight (LOS). It is an imaginary line joining the intersection of the crosshairs of the diaphragm to the optical centre of the objective lens and its continuation.

8. Axis of the telescope:



The axis is an imaginary line passing through the optical centre of the objective lens and the optical centre of the eye-piece.

9. Axis of the level tube:

It is also called the bubble line.

It is a straight line tangential to the longitudinal curve of the level tube at the centre of the tube. It is horizontal when the bubble is in the centre.

10. Vertical axis:

It is the axis about which the telescope can be rotated in the horizontal plane.

11. Horizontal axis:

It is the axis about which the telescope can be rotated in the vertical plane.

It is also called the trunnion axis.

5. ***Adjustment of a Theodolite***

The adjustments of a theodolite are of two kinds:

- Permanent Adjustments
- Temporary Adjustments

Permanent Adjustments

The permanent adjustments are made to establish the relationship between the fundamental lines of the theodolite and once made, they last for a long time. They are essential for the accuracy of observations.

The permanent adjustments in case of a transit theodolites are:

- I. Adjustment of Horizontal Plate Levels. The axis of the plate levels must be perpendicular to the vertical axis.
- II. Collimation Adjustment. The line of collimation should coincide with the axis of the telescope and the axis of the objective slide and should be at right angles to the horizontal axis.
- III. Horizontal axis adjustment. The horizontal axis must be perpendicular to the vertical axis.
- IV. Adjustment of Telescope Level or the Altitude Level Plate Levels. The axis of the telescope levels or the altitude level must be parallel to the line of collimation.
- V. Vertical Circle Index Adjustment. The vertical circle vernier must read zero when the line of collimation is horizontal.

Temporary Adjustments

The temporary adjustments are made at each setup of the instrument before we start taking observations with the instrument. There are three temporary adjustments of a theodolite:

- I. Centering.
- II. Levelling
- III. Focusing

6. *Errors in Theodolite*

INSTRUMENTAL

- Imperfect level of plate level
- LOS (line of sight) not perpendicular to horizontal axis
- Horizontal axis not perpendicular to vertical axis
- LOS & axis of telescope not parallel
- Eccentricity of inner & outer axis
- Imperfect graduation
- Eccentricity of vernier

PERSONAL

- Errors in Manipulation
- Inaccurate centering
- Inaccurate leveling
- Slip
- Wrong tangent screw
- Errors in Sighting & Reading
- Inaccurate sighting
- Ranging rod is not vertical
- Parallax

NATURAL

- Unequal expansion due to temperature change,
- Atmospheric refraction due to high temperature,
- Strong winds,
- Unequal settlement.

7. *How to Use Theodolites*

Theodolites are precision instruments designed to measure horizontal and vertical angles in surveying applications. These instruments come in digital and analog options and use many complex parts to achieve accurate readings. Follow these steps to set up your theodolite and prepare for accurate readings.

Mark the Point

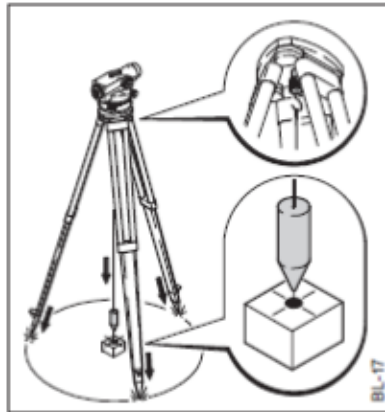
Start with the point where you will set the theodolite. The point needs to be accurately marked, as this will form the basis for measuring distances and angles.

Consider using a surveyor's nail or stake to mark the point. This will ensure you keep an accurate, stable point as you move on to other steps in the process.

Set Up the tripod

The tripod needs to be stable and adjusted to the correct height. Once correctly adjusted, the theodolite should be at eye level. This needs to be adjusted carefully to keep it stable and keep the mounting plate centered on the stake or nail.

In some cases, the point you choose to place the theodolite may need to be adjusted to make room for the tripod. Be sure you have a spot cleared where you can place a tripod without worrying about it tilting or being placed on unlevel ground.



Mount the Theodolite

Once the tripod is firmly set, it's time to place the theodolite on it. Dropping your theodolite can cause permanent damage, so always make sure the tripod is firmly in position before you attempt to mount the instrument. Most theodolites come with screws and mounting knobs. Verify that your particular instrument and tripod can easily connect, or you may not have a stable surface as you start to use it. Follow the instructions carefully for a firm attachment, since tripods and instruments have differing mounting brackets and knobs.

Measure the Height

Before you can start taking accurate measurements, you need to take a reference distance. Measure from the ground to the theodolite. This distance will serve as a reference to other stations as you use the instrument.

Level the Theodolite

Your instrument should have a built-in spirit level. Using the bulls-eye level, check to see if your tripod legs are properly adjusted. Make adjustments as necessary using the legs and the leveling knobs until it's properly positioned.

If you aren't able to firmly place and measure the instrument, then you may need to select a new position. When this occurs, you may need to repeat the initial steps to ensure the correct height and theodolite setup.

Adjust the Small Sight

The vertical plummet, also known as the small sight, is on the bottom of the instrument. It's important that this sight is adjusted to remain over the surveyor's nail or stake. There should be knobs on the bottom of the instrument to allow you to make fine-tuning adjustments until the small sight is in the correct position.

Aim the Crosshairs

Finally, it's time to aim the crosshairs of the main scope. Point at the spot you wish to measure and lock the instrument in place using the locking knobs. Take the vertical and horizontal angle measurements using the viewing scope. This should be on the side of the instrument. These steps complete the theodolite survey. Choose a theodolite for more accurate angle measurements than a transit level. These survey steps are commonly used for building corners or identifying lot lines, but they can also be used to form panels, align frames or for navigation purposes.

C. TOWER VERTICALITY TESTING

When testing for verticality we need to have a tangible value in order to establish whether the tower is vertical or whether the tower is not vertical and adjustments need to be made.

Our towers will not always be perfectly level. However, we will have an allowed deviation to guide us in testing whether a tower is vertical within an accepted tolerance. We will call this accepted deviation the permissible tilt.

1. *Permissible Tilt*

A commonly accepted tolerance for verticality in many engineering and construction standards is 0,25% of the tower's height.

The permissible tilt is equal to 0,25% of the tower's height (H) in metres.

We will use the following formula to calculate this:

$$\text{Permissible Tilt} = 0.0025 \times H$$

Let's calculate the permissible tilt for a 15m tower:

$$\text{Permissible Tilt} = 0.0025 \times H$$

$$\text{Permissible Tilt} = 0.0025 \times 15\text{m}$$

$$\text{Permissible Tilt} = 0.0375\text{m}$$

$$0.0375 = 37,5\text{mm}.$$

We can round this up to 38mm.

Thus, the permissible tilt for a 15m tower is 38mm.

See the table on the following page for permissible tilt values at different elevations.

PERMISSIBLE TILT AT TOWER ELEVATIONS WITH RESPECT TO BASE	
TOWER HEIGHT IN METRES	PERMISSIBLE TILT IN MILLIMETRES
100	250
90	225
80	200
70	175
60	150
50	125
40	100
30	75
27	68
24	60
21	53
18	45
15	38
12	30
9	23

2. *Procedure: Testing Tower Verticality for 3-Leg and 4-Leg towers*

Step 1: Preparation

Ensure that the Base is Level:

- Verify that the tower's base is level using a spirit level or other levelling tools.
- Adjust the base bolts if necessary to level the foundation.
- **It is extremely important to level the base as the verticality test will be inaccurate if the base is not level.**
- Mark the tower legs in a sequence from 1-3 for 3-leg towers or 1-4 for 4-leg towers. It is not important which leg is which number but it is important to mark the legs or note which leg is which and that they follow sequence.

Set Up the Theodolite:

- Mark points A, B, C (&D).
- Try to mark the points at the same distance away from the tower.
- Point A will be midway between legs 1 and 2; Point B will be midway between legs 2 and 3; etc.
- Position the theodolite and tripod at one of the observation points (Point A, B, C, or D). Ensure that you are far enough away from the tower to take accurate readings.
- Ensure that you mark all the points so that you can retake/confirm your measurements later. You can use a sharp object like a thumbnail to stick into soft surfaces. You can use a weighted object like a coin to place/stick onto hard surfaces.
- Level the theodolite carefully using its built-in spirit level.

[Follow step 2A for 3-leg towers or step 2B for 4-leg towers]

Step 2A: Measurement (for 3-Leg Towers)

Point A (Between Legs 1 and 2):

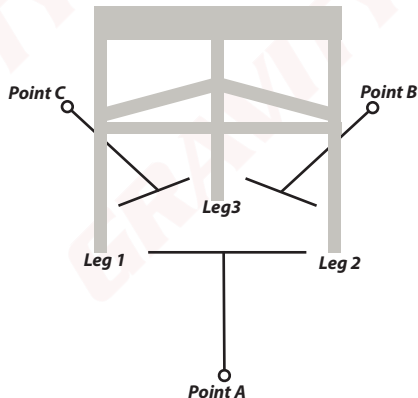
- Position the theodolite at Point A, midway between legs 1 and 2.
- Measure the deviation for leg 1 by aligning the theodolite to a comfortable fixed height (e.g., at shoulder level).
- Measure the deviation for leg 2 at the same height.
- Use a tape measure to determine the horizontal difference between the reference line (theodolite reading) and the actual leg position (in millimetres).

Point B (Between Legs 2 and 3):

- Move the theodolite to Point B, midway between legs 2 and 3.
- Measure the deviations for legs 2 and 3 using the same method.

Point C (Between Legs 3 and 1):

- Move the theodolite to Point C, midway between legs 3 and 1.
- Measure the deviations for legs 3 and 1.



[Follow step 2A for 3-leg towers or step 2B for 4-leg towers]

Step 2B: Measurement (for 4-Leg Towers)

Point A (Between Legs 1 and 2):

- Position the theodolite at Point A, midway between legs 1 and 2.
- Measure the deviation for leg 1 and then leg 2 at the same height.

Point B (Between Legs 2 and 3):

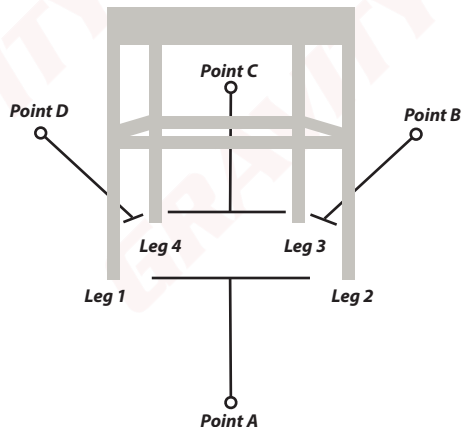
- Move the theodolite to Point B, midway between legs 2 and 3.
- Measure the deviations for legs 2 and 3.

Point C (Between Legs 3 and 4):

- Move the theodolite to Point C, midway between legs 3 and 4.
- Measure the deviations for legs 3 and 4.

Point D (Between Legs 4 and 1):

- Move the theodolite to Point D, midway between legs 4 and 1.
- Measure the deviations for legs 4 and 1.



Step 3: Compare Deviations

Calculate Permissible Tilt:

- Use the formula $\Delta \leq 0.0025 \times H$, where Δ is the Deviation (which should be equal to or less than the permissible tilt) and H is the height of the tower (in metres).
- Remember to convert the permissible tilt to millimetres when comparing deviations.

Compare Each Measured Deviation:

- Measure the horizontal deviations (in millimetres) for each leg using a tape measure.
- Check if the measured deviations (in millimetres) for each leg are less than or equal to the permissible tilt:
 - » If $\Delta \leq$ Permissible Tilt, the leg is within acceptable limits.
 - » If $\Delta >$ Permissible Tilt, adjustments are required.

Step 4: Adjust and Repeat

Adjust Base Bolts:

- Tighten or loosen the bolts at the base of the tower legs to correct any deviations exceeding the permissible tilt.

Recheck Measurements:

- Repeat the process for all legs until all deviations are within permissible limits.

Step 5: Final Confirmation

- Perform one last set of measurements across all legs to confirm that the tower is vertical.
Document the results, ensuring all measured deviations meet the permissible tilt standard.

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